

08L2-Feature Extraction

Generated by Scribe.AI

December 18, 2024

Slide 1

Content

CS 24300 - AI Basics

Tony Bergstrom

Slides adapted from Yexiang Xue

Description

This slide is a title slide for a lecture on Artificial Intelligence Basics, likely part of a course called CS 24300. It indicates the course title, the instructor's name (Tony Bergstrom), and the source of the slides (Yexiang Xue). This is a standard introductory slide, providing basic course information.

Figures

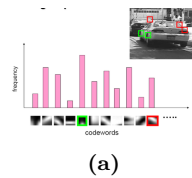
Slide 2

Content

Description

This slide, likely part of a lecture on Artificial Intelligence Basics, introduces the concept of Bag of Features (BoF) for object recognition. A histogram of frequencies (in pink) is shown, representing the distribution of codewords. A grayscale image of a street scene with a car is displayed, along with several bounding boxes (red and green squares) highlighting detected objects. The codewords, represented by various grayscale patterns, are shown below the histogram. The context suggests a discussion of image feature extraction, where codewords are extracted from images and used to represent objects. The histogram visualizes the frequency of occurrence of these codewords, and the bounding boxes on the image show the results of object detection using the BoF approach. The slide likely explains how the codewords are generated and used to classify objects in images.

Figures

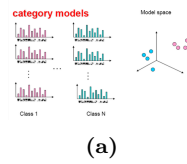


Slide 3

Content

Description

Figures



Slide 4

Content

Edge Detection

Description

This slide displays a title, "Edge Detection," likely indicating the topic of the following slides in a lecture on Artificial Intelligence Basics. No further content is present on the slide.

Figures

Slide 5

Content

Why edges?

Resilience to lighting and color

- useful for recognition, matching patches across images

Description

This slide, likely part of a lecture on image processing or computer vision within an Artificial Intelligence Basics course, discusses the advantages of using edges in image analysis. The title "Why edges?" indicates the focus of the slide. The text below the title explains that edge detection is resilient to variations in lighting and color, making it a valuable tool for object recognition. The bullet point emphasizes that edges are useful for identifying and matching similar features (patches) across different images, even if the lighting or color of the images differs. The slide likely contains images illustrating the concept, but the provided text does not include them. The context suggests a discussion of image feature extraction and matching techniques.

Figures



(a)



(b)



(c)



(d)



(e)

Slide 6

Content

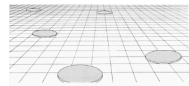
Description

This slide, likely part of a lecture on image processing or computer vision within an Artificial Intelligence Basics course, discusses the importance of edges in image analysis. The title "Why edges?" introduces the topic. The text below the title explains that edges provide cues for shape and geometry. The bullet point emphasizes that edges are useful for object recognition and understanding the three-dimensional structure of objects in an image. The slide includes two illustrative images. One image, credited to Attneave, shows a simplified, line-drawing representation of an object (likely a cat), highlighting its edges. The other image, credited to Jitendra Malik, shows a perspective drawing of circles on a grid, illustrating how edges can be used to perceive depth and three-dimensional structure. The context suggests a discussion of image feature extraction and how edges are crucial for understanding the spatial relationships within an image.

Figures



(a)



(b)

Slide 7

Content

Why edges?

Grouping pixels into objects ("perceptual organization")

Description

This slide, likely part of a lecture on image processing or computer vision within an Artificial Intelligence Basics course, discusses the importance of edges in image analysis. The title "Why edges?" introduces the topic. The text below the title explains that edge detection is crucial for grouping pixels into meaningful objects, a process often referred to as "perceptual organization." The slide contains two images. The image on the left is a photograph of a tiger in its natural habitat. The image on the right is a simplified representation of the tiger, highlighting only the edges of the tiger's shape against the background. The arrow between the images indicates the transformation from a complex image to a simplified representation based on edges. The context suggests a discussion of image segmentation and feature extraction techniques, where edges are used to identify and delineate objects within an image.

Figures



(a)

Slide 8

Content

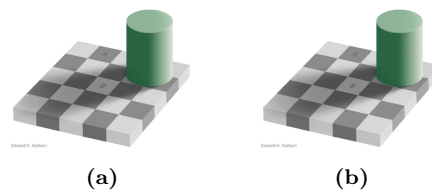
Edges

Edges are curves in the image, across which the brightness changes "a lot"
Corners/Junctions

Description

This slide, likely part of a lecture on image processing or computer vision within an Artificial Intelligence Basics course, defines "edges." The title "Edges" introduces the topic. The text below the title states that edges are curves in an image where the brightness changes significantly. The slide also includes the label "Corners/Junctions," suggesting that this is a related concept. A 3D illustration of a cylinder resting on a checkered surface is present. The labels "A" and "B" are positioned on the checkered surface, likely to indicate points of interest or reference for the edges. The image is credited to Edward H. Adelson. The context suggests a discussion of image feature extraction, specifically how edges and corners are used to identify and describe objects in an image.

Figures



Slide 9

Content

Closeup of edges

Description

This slide, likely part of a lecture on image processing or computer vision within an Artificial Intelligence Basics course, focuses on the concept of edges in images. The title "Closeup of edges" indicates the slide's subject. The slide contains a high-resolution image of a section of a building, likely a palace or government building, with a large arched doorway. The image is a close-up, emphasizing the details of the building's architecture. The context suggests a discussion of edge detection techniques in image analysis, where the sharp transitions in brightness and color between different parts of the building (e.g., the archway, walls, and ground) are considered edges. The slide is likely to be followed by a discussion of how these edges can be extracted and used for tasks like object recognition, image segmentation, or 3D reconstruction. The source is credited as "D. Hoiem," implying a specific researcher or textbook in the field of computer vision.

Figures



(a)

Slide 10

Content

Closeup of edges

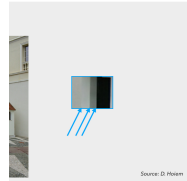
Description

This slide, likely part of a lecture on image processing or computer vision within an Artificial Intelligence Basics course, displays a close-up image of a section of a building, likely a palace or government building, with a large arched doorway. The title "Closeup of edges" indicates the focus on the sharp transitions in brightness and color between different parts of the building, which are considered edges. The image emphasizes the details of the building's architecture, highlighting the edges of the archway, walls, and ground. Three light-blue arrows point from a small, highlighted rectangular area (likely a cropped portion of the image) to the main image. This suggests a comparison or analysis of the highlighted area's edges with the larger image's edges. The source is credited as "D. Hoiem," implying a specific researcher or textbook in the field of computer vision. The context suggests a discussion of edge detection techniques in image analysis, where the sharp transitions in brightness and color are used to identify and delineate objects within an image. The slide is likely to be followed by a discussion of how these edges can be extracted and used for tasks like object recognition, image segmentation, or 3D reconstruction.

Figures



(a)



(b)

Slide 11

Content

Closeup of edges

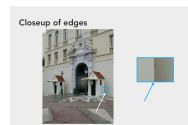
Description

This slide, likely part of a lecture on image processing or computer vision within an Artificial Intelligence Basics course, displays a title "Closeup of edges." A full-color image of a section of a building, likely a palace or government building, is shown. The image is a close-up, focusing on the details of the building's architecture, specifically a large arched doorway. A smaller, rectangular, light-blue box is drawn around a portion of the building's facade. A light-blue arrow points from the smaller box to a portion of the larger image. The context suggests a discussion of edge detection techniques in image analysis. The sharp transitions in brightness and color between different parts of the building (e.g., the archway, walls, and ground) are considered edges. The slide is likely to be followed by a discussion of how these edges can be extracted and used for tasks like object recognition, image segmentation, or 3D reconstruction. The source is credited as "D. Hoiem," implying a specific researcher or textbook in the field of computer vision. The slide is designed to illustrate the concept of edges in an image and how they can be highlighted or analyzed in detail.

Figures



(a)



(b)

Slide 12

Content

Closeup of edges

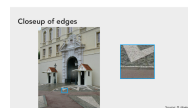
Description

This slide, part of an Artificial Intelligence Basics course likely focusing on image processing or computer vision, displays a title "Closeup of edges." A full-color image of a section of a building, likely a palace or government building, is shown. The image is a close-up, focusing on the details of the building's architecture, specifically a large arched doorway. A smaller, rectangular, light-blue box is drawn around a portion of the building's facade, specifically the pavement in front of the archway. A light-blue arrow points from the smaller box to a portion of the larger image, likely to highlight a specific area of interest within the larger image. The context suggests a discussion of edge detection techniques in image analysis. The sharp transitions in brightness and color between different parts of the building (e.g., the archway, walls, and ground) and the pavement are considered edges. The slide is likely to be followed by a discussion of how these edges can be extracted and used for tasks like object recognition, image segmentation, or 3D reconstruction. The image on the right, cropped from the main image, is a close-up of the pavement, highlighting the texture and pattern of the stones. The source is credited as "D. Hoiem," implying a specific researcher or textbook in the field of computer vision. The slide is designed to illustrate the concept of edges in an image and how they can be highlighted or analyzed in detail, potentially comparing the edges of a smaller section to the larger image.

Figures



(a)



(b)

Slide 13

Content

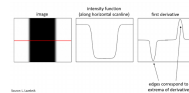
Characterizing edges

An edge is a place of rapid change in the image intensity function

Description

This slide, part of an Artificial Intelligence Basics course likely focusing on image processing or computer vision, defines an edge. It states that an edge is a location where there is a rapid change in the image intensity function. The slide visually illustrates this concept with three panels. The leftmost panel shows a grayscale image with a sharp transition from dark to light, representing an edge. A horizontal red line is drawn across the image, indicating a scanline. The middle panel displays a graph of an intensity function (likely a plot of intensity values along the scanline). The graph shows a smooth transition, representing the gradual change in intensity values. The rightmost panel shows a graph of the first derivative of the intensity function. This graph exhibits a sharp peak (or trough) where the intensity changes rapidly, corresponding to the edge in the image. A small arrow points from the peak in the derivative graph to the edge in the image. The text "edges correspond to extrema of derivative" is included, highlighting the relationship between the location of the edge in the image and the extrema (maximum or minimum) of the derivative graph. The source is credited as "L. Lazebnik," indicating a specific researcher or textbook in the field of computer vision. The slide is designed to illustrate the mathematical concept of edge detection, where edges are identified by analyzing the rate of change in intensity values.

Figures



(a)

Slide 14

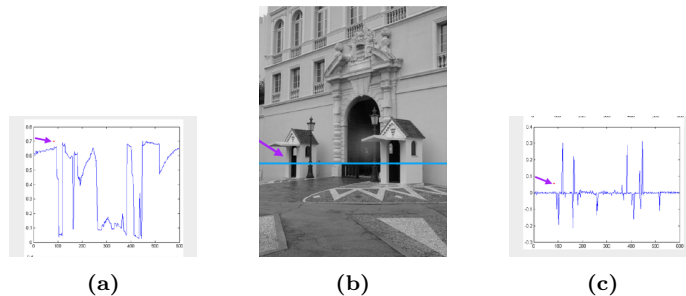
Content

Intensity profile

Description

This slide, likely part of a lecture on image processing or computer vision within an Artificial Intelligence Basics course, displays a title "Intensity profile." A grayscale image of a section of a building, likely a palace or government building, is shown. A horizontal light-blue line is drawn across the image, indicating a scanline. A purple arrow points from a portion of the image to the horizontal line, indicating the scanline. Two graphs are displayed below the image. The top graph plots intensity values against position along the scanline. The bottom graph plots the first derivative of the intensity values against position along the scanline. The graphs show the intensity profile along the horizontal line in the image. The intensity profile is a plot of the intensity values along a scanline in the image. The intensity profile is used to analyze the changes in intensity along the scanline, which can be used to detect edges and other features in the image. The source is credited as "D. Hoiem," implying a specific researcher or textbook in the field of computer vision. The slide is designed to illustrate the concept of intensity profiles and how they can be used to analyze images.

Figures



Slide 15

Content

Image derivatives

How can we differentiate a digital image $F[x, y]$?

- We can take a discrete derivative! (Finite difference)
- $\frac{\partial f}{\partial x}[x, y] \approx F[x + 1, y] - F[x, y]$
- How would you implement this as a linear filter?
- $H_x = \begin{bmatrix} -1 \\ +1 \end{bmatrix}$
- $H_y = \begin{bmatrix} -1 \\ +1 \end{bmatrix}$

Description

This slide, part of an Artificial Intelligence Basics course likely focusing on image processing or computer vision, discusses how to calculate derivatives of digital images. The title is "Image derivatives." The slide explains that discrete derivatives can be calculated using finite differences. The formula $\frac{\partial f}{\partial x}[x, y] \approx F[x + 1, y] - F[x, y]$ is presented, showing how the partial derivative with respect to x can be approximated by the difference in pixel values between adjacent pixels in the x-direction. The slide then asks how to implement this as a linear filter. Two linear filters, H_x and H_y , are presented as matrices. H_x is a 2x1 matrix with elements $[-1, +1]$, and H_y is a 2x1 matrix with elements $[-1, +1]$. The context suggests that these filters can be used to perform the discrete derivative operation on an image. The source is credited as "S. Seitz," indicating a specific researcher or textbook in the field of computer vision.

Figures

Slide 16

Content

Description

This slide, from an Artificial Intelligence Basics course likely covering image processing or computer vision, demonstrates a 2D moving average operation. The title is "Moving Average In 2D." Two matrices, $F[x, y]$ and $G[x, y]$, are displayed. $F[x, y]$ represents an input image, likely a grayscale image, where each element represents a pixel intensity value. The values are all 90 except for a few 0s. The matrix $G[x, y]$ is the output of the moving average operation. The notation $*[-1 \ +1]$ above the matrix suggests a filter or kernel being applied to the input image to compute the moving average. The red-highlighted elements in the $F[x, y]$ matrix indicate the region of the image being considered for the moving average calculation at a specific location. The moving average operation calculates the average intensity value of a neighborhood of pixels in the input image to produce the output image. The source is credited as "S. Seitz," indicating a specific researcher or textbook in the field of computer vision.

Figures

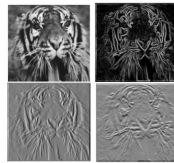
Slide 17

Content

Description

This slide, from an Artificial Intelligence Basics course likely covering image processing or computer vision, introduces the concept of the image gradient. The title is "Image Gradient." The slide displays a formula for calculating the magnitude of the gradient of an image function $f(x, y)$. The formula is: $||\nabla f|| = \sqrt{\left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2}$. This formula calculates the magnitude of the gradient vector, which represents the rate of change of the image intensity in different directions. The slide also shows four grayscale images. The top row shows the original image of a tiger's face. The bottom row shows the results of applying filters to calculate the partial derivatives of the image with respect to x and y. The images in the bottom row represent the gradient in the x-direction and the y-direction, respectively. The source is credited as "L. Lazebnik," indicating a specific researcher or textbook in the field of computer vision. The slide is designed to illustrate the concept of image gradients and how they can be used to analyze images, likely for tasks like edge detection or feature extraction.

Figures



(a)

Slide 18

Content

Generalized: edge filters of different directions

Description

This slide, part of an Artificial Intelligence Basics course likely focusing on image processing or computer vision, displays a set of grayscale images. The title, "Generalized: edge filters of different directions," indicates that these images represent different edge filters, each designed to detect edges oriented in a specific direction. The slide shows several small grayscale images arranged in a row. Each image appears to be a small, elongated, dark-light-dark pattern, suggesting a filter kernel designed to detect edges. The filters are likely oriented in different directions, with the filters showing different orientations. The context of the course suggests that these filters are used in image processing techniques to identify edges in images, which are crucial features for tasks like object recognition or image segmentation. The filters are likely part of a larger discussion on edge detection or feature extraction techniques in image analysis.

Figures



(a)

Slide 19

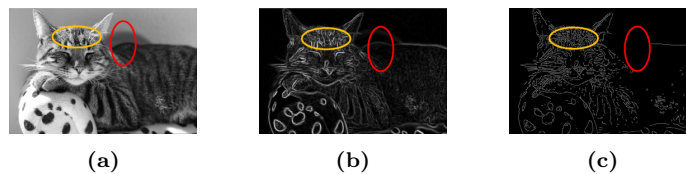
Content

Canny edge detector

Description

This slide, from an Artificial Intelligence Basics course likely covering image processing or computer vision, demonstrates the output of a Canny edge detector applied to an image of a gray tabby cat. The title is "Canny edge detector." The slide displays three images. The top image is the original grayscale image of the cat. The middle and bottom images are the results of applying the Canny edge detector to the original image. The Canny edge detector is an algorithm designed to detect edges in images. The algorithm typically involves several steps, including smoothing the image, calculating the gradient magnitude and direction, non-maximum suppression, and hysteresis thresholding. The output of the Canny edge detector is a binary image where edges are highlighted with white or light gray pixels. Yellow circles are drawn around the detected edges in the cat's head region, and red circles are drawn around the detected edges in the cat's shoulder region. The purpose of the slide is to illustrate the output of the Canny edge detector and how it highlights the edges of the cat's head and shoulder. The context of the course suggests that this slide is part of a discussion on edge detection techniques in image analysis.

Figures



Slide 20

Content

Corner detection

Main idea: Translating window should cause large differences in patch appearance

Description

This slide, part of an Artificial Intelligence Basics course likely focusing on computer vision or image processing, introduces the concept of corner detection. The title is "Corner detection." The main idea, presented as a statement, is that a small window (represented by the smaller squares) translated over an image should show significant differences in the appearance of the patch (the larger blue square) if a corner is present. The slide uses a visual representation of a larger blue square with smaller squares (in different colors) positioned around it. The smaller squares represent a "window" that is being translated. The idea is that if the window is moved slightly, the appearance of the patch within the window will change significantly at a corner, whereas it will change less at a smooth region. This is a fundamental concept in corner detection algorithms, which aim to identify locations in an image where the image intensity changes significantly in multiple directions. The context of the course suggests that this slide is part of a discussion on feature extraction techniques in image analysis, specifically focusing on corners as important features.

Figures

Slide 21

Content

Announcements

Midterm: In class Thurs, 17 Oct

Final: Wed, 11 Dec, 10:30 am - 12:30 pm, BRNG 2280

Lab - manual grading

Homework 2: to be released soon.

Description

This slide displays announcements for an Artificial Intelligence Basics course. It lists the date and time for the midterm exam (Thursday, October 17th), the final exam (Wednesday, December 11th, from 10:30 AM to 12:30 PM in BRNG 2280), and indicates that labs will be manually graded. Finally, it announces that homework assignment 2 will be released soon. The context of the slide is clearly related to course scheduling and administrative information.

Figures

Slide 22

Content

Corner detection

Main idea: Translating window should cause large differences in patch appearance

Description

This slide, part of an Artificial Intelligence Basics course likely covering computer vision or image processing, introduces the concept of corner detection. The title is "Corner detection." The main idea is presented as a concise statement: "Translating window should cause large differences in patch appearance." A visual representation accompanies this statement. A larger, medium-blue rectangle is shown, with smaller, differently colored (light-green and dark-purple) squares positioned around it. These smaller squares represent a "window" that is being translated. The idea is that if the window is moved slightly, the appearance of the patch within the window will change significantly at a corner, whereas it will change less at a smooth region. This visual aids in understanding the core principle behind corner detection algorithms, which aim to identify locations in an image where the image intensity changes significantly in multiple directions. The context of the course suggests that this slide is part of a discussion on feature extraction techniques in image analysis, specifically focusing on corners as important features.

Figures

Slide 23

Content

Corner Detection: Basic Idea

We should easily recognize the point by looking through a small window
Shifting a window in any direction should give a large change in intensity

flat region:

no change in all
directions

edge:

no change along the
edge direction

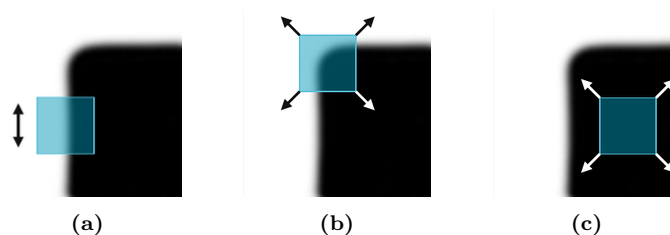
corner:

significant change in all
directions

Description

This slide, part of an Artificial Intelligence Basics course likely focusing on computer vision or image processing, explains the fundamental concept behind corner detection. The title is "Corner Detection: Basic Idea." The slide illustrates the idea using three visual examples. A small, light-teal colored square represents a "window" that is moved (or translated) over an image. The first image shows a "flat region" where the intensity of the image within the window does not change significantly when the window is shifted in any direction. The second image shows an "edge" where the intensity changes significantly only when the window is shifted perpendicular to the edge, but not when shifted along the edge. The third image shows a "corner" where the intensity changes significantly when the window is shifted in any direction. The text below each image describes the corresponding feature: "flat region," "edge," and "corner." The key takeaway is that corner detection algorithms identify locations in an image where the image intensity changes significantly in multiple directions. The slide is likely part of a discussion on feature extraction techniques in image analysis, specifically focusing on corners as important features. The source is attributed to A. Efros.

Figures



Slide 24

Content

Corner detection the math

Consider shifting the window W by (u, v)

How do the pixels in W change?

$$\phi_0 = [I(0, 0), I(0, 1), \dots, I(n, n)]$$

$$\phi_1 = [I(0 + u, 0 + v), I(0 + u, 1 + v), \dots, I(n + u, n + v)]$$

Write pixels in window as a vector:

$$E(u, v) = \|\phi_0 - \phi_1\|^2 = \sum_{(x, y) \in W} (I(x, y) - I(x + u, y + v))^2$$

Description

This slide, part of an Artificial Intelligence Basics course likely covering computer vision or image processing, details the mathematical formulation for corner detection. The title is "Corner detection the math." The slide describes how to calculate the change in pixel values when a window W is shifted by a displacement (u, v) . The notation $I(x, y)$ represents the intensity value of the pixel at coordinates (x, y) . ϕ_0 represents the vector of pixel intensities within the window in its original position, and ϕ_1 represents the vector of pixel intensities after the window is shifted by (u, v) . The equation $E(u, v) = \|\phi_0 - \phi_1\|^2$ calculates the difference between the two vectors, effectively measuring the change in pixel intensities. The summation term $\sum_{(x, y) \in W} (I(x, y) - I(x + u, y + v))^2$ quantifies this difference by summing the squared differences of corresponding pixel intensities across the window. A diagram is included, showing a window W and its shifted position, illustrating the concept of the displacement (u, v) . The context of the course suggests that this slide is part of a discussion on feature extraction techniques in image analysis, specifically focusing on corner detection algorithms. The goal is to find locations in an image where the intensity changes significantly in multiple directions when the window is shifted, which is a key characteristic of corners.

Figures

Slide 25

Content

Corner detection: the math

Consider shifting the window W by (u, v)

How do the pixels in W change?

- Compare each pixel before and after by summing up the squared differences (SSD)
- Define an SSD "error":

$$E(u, v) = \sum_{(x, y) \in W} [I(x + u, y + v) - I(x, y)]^2$$

We want $E(u, v)$ to be as high as possible for all u, v to indicate a corner.

Description

This slide, part of an Artificial Intelligence Basics course likely covering computer vision or image processing, details the mathematical approach to corner detection. The title is "Corner detection: the math." The slide describes how to quantify the change in pixel intensities when a window W is shifted by a displacement (u, v) . The notation $I(x, y)$ represents the intensity value of the pixel at coordinates (x, y) . The key concept is to compare the pixel intensities before and after the shift using the sum of squared differences (SSD).

The equation $E(u, v) = \sum_{(x, y) \in W} [I(x + u, y + v) - I(x, y)]^2$ calculates the total squared difference across all

pixels within the window W after the shift. A diagram is included, showing a window W and its shifted position, illustrating the concept of the displacement (u, v) . The slide emphasizes that a high value of $E(u, v)$ for all possible shifts (u, v) indicates a corner. The context of the course suggests that this slide is part of a discussion on feature extraction techniques in image analysis, specifically focusing on corner detection algorithms. The goal is to find locations in an image where the intensity changes significantly in multiple directions when the window is shifted, which is a key characteristic of corners.

Figures

Slide 26

Content

Corner Detection: Mathematics

Change in appearance of window $w(x, y)$ for the shift $[u, v]$:

$$E(u, v) = \sum_{x, y} w(x, y) [I(x + u, y + v) - I(x, y)]^2$$

Description

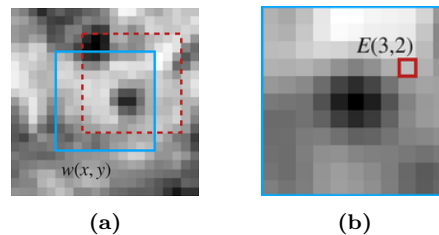
This slide, part of an Artificial Intelligence Basics course likely focusing on computer vision or image processing, presents the mathematical formula for corner detection. The title is "Corner Detection: Mathematics."

The slide shows the equation $E(u, v) = \sum_{x, y} w(x, y) [I(x + u, y + v) - I(x, y)]^2$. This equation calculates

the change in appearance of a window $w(x, y)$ in an image when shifted by a displacement (u, v) . $w(x, y)$ represents a weighting function applied to the pixels within the window. $I(x, y)$ represents the intensity value of the pixel at coordinates (x, y) . The summation $\sum_{x, y}$ indicates that the squared difference between

the pixel intensities at the shifted and original positions is calculated for all pixels within the window and then summed up. The slide also includes a visual representation of a window $w(x, y)$ in an image, with a dashed red box showing a shifted position of the window. A small square labeled $E(3, 2)$ is shown in the image, indicating a specific calculation of the change in appearance for a shift of 3 in the x-direction and 2 in the y-direction. The context of the course suggests that this slide is part of a discussion on feature extraction techniques in image analysis, specifically focusing on corner detection algorithms. The goal is to find locations in an image where the intensity changes significantly in multiple directions when the window is shifted, which is a key characteristic of corners.

Figures



Slide 27

Content

Description

This slide, part of an Artificial Intelligence Basics course likely covering computer vision or image processing, presents the mathematical formulation for corner detection. The title is "Corner Detection: Mathematics." The slide shows the equation $E(u, v) = \sum_{x, y} w(x, y) [I(x + u, y + v) - I(x, y)]^2$, which calculates the change

in appearance of a window $w(x, y)$ in an image when shifted by a displacement (u, v) . $w(x, y)$ is a window function, and $I(x, y)$ represents the intensity value of the pixel at coordinates (x, y) . The equation sums the squared differences between the pixel intensities at the shifted and original positions, weighted by the window function. The slide also shows two examples of window functions: a rectangular function (1 inside the window, 0 outside) and a Gaussian function. The slide emphasizes that a corner is detected if the minimum value of $E(u, v)$ across all possible shifts (u, v) is large. The slide concludes by mentioning that optimizing for the minimum $E(u, v)$ is a Singular Value Decomposition (SVD) problem, but the method for solving it is not covered in this lecture. The context of the course suggests that this slide is part of a discussion on feature extraction techniques in image analysis, specifically focusing on corner detection algorithms. The goal is to find locations in an image where the intensity changes significantly in multiple directions when the window is shifted, which is a key characteristic of corners.

Figures

Slide 28

Content

Harris detector example

Description

This slide, part of an Artificial Intelligence Basics course likely covering computer vision or image processing, displays an example of the Harris corner detector. The slide title is "Harris detector example." The content consists of an image, likely a digital photograph or graphic, showing two cartoon-like cows. The cows are depicted in a style that suggests a simple, stylized representation of a cow, with black and white spots and a long, rope-like neck. The image is presented in a way that suggests the output of a corner detection algorithm, where the corners of the image are highlighted or emphasized. The context of the course suggests that this slide is part of a discussion on feature extraction techniques in image analysis, specifically focusing on corner detection algorithms. The goal is to demonstrate how the Harris corner detector identifies corners in an image, likely by highlighting the corners of the cows' bodies and heads. The image is a visual aid to illustrate the concept of corner detection, rather than a complex mathematical formula or algorithm description.

Figures



(a)

Slide 29

Content

Harris features (in red)

Description

This slide, part of an Artificial Intelligence Basics course likely covering computer vision, displays an image of a stylized cartoon cow. The image shows two identical cartoon cows, side-by-side. Small red dots are overlaid on the image, marking detected features. The title, "Harris features (in red)," indicates that the red dots represent features detected by the Harris corner detector algorithm. The context suggests that the slide is illustrating the output of a corner detection algorithm, highlighting prominent corners and potentially other significant image features. The red dots are likely placed at locations where the algorithm identified significant changes in image intensity or gradient, which are characteristic of corners or other important features in an image. The cartoon style of the image is likely used for simplicity and clarity in demonstrating the algorithm's output, rather than for any specific analysis of the cow itself.

Figures



(a)

Slide 30

Content

Putting it together: Vision pyramids

Description

This slide, part of an Artificial Intelligence Basics course likely covering computer vision, discusses the concept of vision pyramids. It describes a method for extracting features from images. The slide shows a series of images, likely of a painting, progressively smaller and blurred. The notation F_0 , F_1 , F_2 , etc., likely represents different levels of the pyramid. The image F_0 is the original image. The image F_1 is a blurred and subsampled version of F_0 . The image F_2 is a blurred and subsampled version of F_1 , and so on. The arrows indicate the process of blurring and subsampling. The notation $F_0 * H$ likely represents the result of applying a filter (e.g., blurring) to F_0 and then subsampling. The text "Filter the image, then subsample" summarizes the process. The slide emphasizes that each pixel in a small image (e.g., F_2) can be a good feature descriptor, summarizing a local region of the original image (F_0). The slide also mentions other possible feature descriptors, such as edge graphs, corner graphs, and color histograms. The context suggests that this slide is part of a discussion on image feature extraction techniques, specifically focusing on the concept of image pyramids and how they can be used to extract features at different scales and resolutions.

Figures



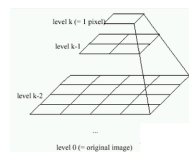
Slide 31

Content

Description

This slide, from an Artificial Intelligence Basics course likely covering computer vision, introduces the concept of Gaussian pyramids. It describes how an image of size $N \times N$ can be represented as a pyramid of progressively smaller images, where each level ($2^k \times 2^k$) is a downsampled version of the previous level. The image shows a graphical representation of this pyramid structure, with levels $k-2$, $k-1$, k , and 0 (the original image) depicted. The levels are progressively smaller, with level k containing only one pixel. The slide also mentions that this concept is related to mip maps, a technique used in computer graphics, and highlights the broad applications of Gaussian pyramids in computer vision. The reference to Burt and Adelson (1983) and Williams (1983) indicates that this is a well-established technique in the field.

Figures



(a)

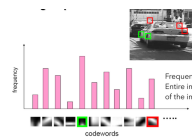
Slide 32

Content

Description

This slide, part of an Artificial Intelligence Basics course likely covering computer vision, describes the "Bag of Features" approach to object recognition. A histogram (pink bars) displays the frequency of different "codewords" (visual features). These codewords are likely extracted from the image using a technique like SIFT or SURF. The slide shows a grayscale image (in the inset) with several colored boxes overlaid. These boxes likely represent regions where specific codewords were found to be frequent. The arrow from the histogram to the text "Fed into classifiers" indicates that the frequency counts of these codewords are used as input to a classifier (e.g., logistic regression or perceptron). The frequency maps are used to determine the presence or absence of objects in the image. The slide emphasizes that these frequency maps can be computed across the entire image or for specific regions within the image. The context suggests that the slide is part of a discussion on image feature extraction and classification techniques.

Figures



(a)

Slide 33

Content

Derivatives and convolution

Differentiation is linear

$$\frac{\partial(af(x) + bg(x))}{\partial x} = a \frac{\partial f(x)}{\partial x} + b \frac{\partial g(x)}{\partial x}$$

Differentiation is shift-invariant: Derivative of shifted signal is shifted derivative Hence, differentiation can be represented as convolution!

Description

This slide, likely from a course on signal processing or a related field within Artificial Intelligence Basics, discusses the relationship between differentiation and convolution. It states that differentiation is a linear operation, meaning the derivative of a sum of functions is the sum of the derivatives of the individual functions. This is shown mathematically with the equation $\frac{\partial(af(x) + bg(x))}{\partial x} = a \frac{\partial f(x)}{\partial x} + b \frac{\partial g(x)}{\partial x}$. The slide also highlights that differentiation is shift-invariant. This means that if a function is shifted, its derivative is also shifted by the same amount. The final statement, "Hence, differentiation can be represented as convolution!", indicates that the properties of linearity and shift-invariance allow differentiation to be expressed using the convolution operation. This is a crucial concept in signal processing, as convolution is often computationally more efficient than direct differentiation, especially when dealing with large datasets or signals. The context suggests that the slide is part of a discussion on signal processing techniques and their applications.

Figures

Slide 34

Content

Image gradient

The gradient points in the direction of most rapid increase in intensity

The gradient of an image: $\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right]$

The edge strength is given by the gradient magnitude:

$$\nabla f = \left| \frac{\partial f}{\partial x} \right| \quad \nabla f = \left| \frac{\partial f}{\partial y} \right| \quad \nabla f = \left| \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right|$$

The gradient direction is given by:

$$\|\nabla f\| = \sqrt{\left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2}$$

The direction of the edge?

$$\theta = \arctan^{-1} \left(\frac{\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial y}} \right)$$

Description

This slide from an Artificial Intelligence Basics course, likely covering computer vision, introduces the concept of image gradients. It defines the gradient of an image as a vector $\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right]$, where $\frac{\partial f}{\partial x}$ and

$\frac{\partial f}{\partial y}$ represent the partial derivatives of the image intensity function f with respect to x and y , respectively. The slide explains that the gradient points in the direction of the most rapid increase in intensity. It also defines the edge strength (magnitude) as the magnitude of the gradient vector, calculated as

$\|\nabla f\| = \sqrt{\left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2}$. The gradient direction (θ) is given by the arctangent of the ratio of the partial

derivatives: $\theta = \arctan^{-1} \left(\frac{\frac{\partial f}{\partial x}}{\frac{\partial f}{\partial y}} \right)$. The slide includes visual representations of the gradient vector, showing its direction and magnitude in relation to the image intensity changes. The context suggests that this slide is part of a discussion on edge detection techniques in image processing, where the gradient information is used to identify and characterize edges in an image.

Figures

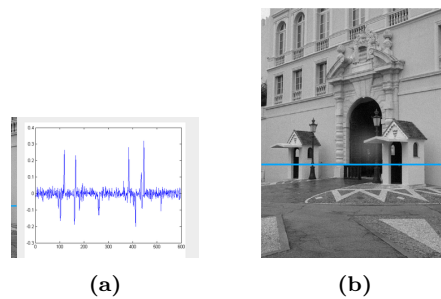
Slide 35

Content

Description

This slide, likely from a computer vision or image processing section of an Artificial Intelligence Basics course, demonstrates the effect of Gaussian noise on an image. The left side of the slide shows a grayscale image (likely a photograph) of a building. The right side displays a plot of a signal, likely representing the intensity values of a line or region within the image. The plot shows a fluctuating signal with small, random variations superimposed on it. The title "With a little Gaussian noise" indicates that the signal or image on the right has been artificially corrupted with Gaussian noise. The noise is characterized by its random nature and the bell-shaped probability distribution of the Gaussian distribution. The horizontal axis of the plot likely represents time or pixel position, and the vertical axis represents the intensity values. The blue horizontal line in the image on the left is likely a reference line or a baseline for comparison. The source is credited to D. Hoiem, suggesting a specific author or researcher associated with the material. The context suggests that the slide is part of a discussion on image noise, its effects on image processing tasks, and techniques for noise reduction.

Figures



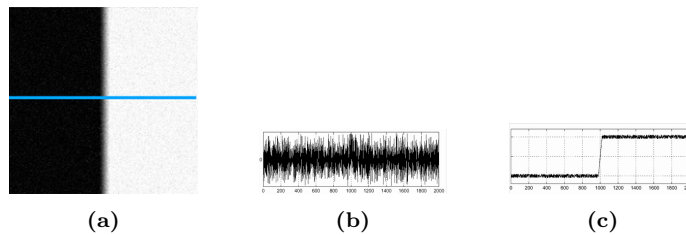
Slide 36

Content

Description

This slide from an Artificial Intelligence Basics course, likely focusing on image processing, demonstrates the effects of noise on an image and its derivative. A noisy grayscale image of a step function (a transition from black to white) is shown on the left. A horizontal blue line represents the approximate location of the edge. The corresponding signal $f(x)$ (likely representing the intensity values along a horizontal scanline of the image) is plotted in the middle. This signal shows a gradual transition from one intensity level to another, but with significant random fluctuations (noise). The plot on the right shows the derivative of the signal, $\frac{d}{dx}f(x)$. This derivative highlights the rapid changes in intensity, which correspond to the edge. However, the noise in the original signal is amplified in the derivative, making it difficult to precisely locate the edge. The question "Where is the edge?" emphasizes the challenge of edge detection in noisy images. The source is credited to S. Seitz, indicating a specific author or researcher associated with the material. The context suggests that the slide is part of a discussion on image noise, its effects on image processing tasks, and techniques for noise reduction, particularly edge detection.

Figures



Slide 37

Content

Effects of noise

Noise is high frequency

Differentiation accentuates noise

$$\frac{d}{dx} \sin(wx) = w \cdot \cos(wx)$$

Description

This slide from an Artificial Intelligence Basics course, likely focusing on image processing or signal processing, discusses the effects of noise on signals and how differentiation amplifies noise. The title "Effects of noise" clearly indicates the topic. The text states that noise is characterized as having a high frequency. Crucially, it explains that differentiating a signal (taking its derivative) will amplify the noise component.

The mathematical example $\frac{d}{dx} \sin(wx) = w \cdot \cos(wx)$ demonstrates the relationship between the derivative of a sinusoidal function and its amplitude. This example is likely used to illustrate how noise, which often has a higher frequency component than the signal, becomes more prominent when the signal is differentiated. The context suggests that the slide is part of a discussion on image noise, its effects on image processing tasks, and techniques for noise reduction, particularly edge detection.

Figures

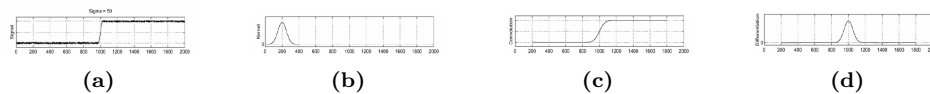
Slide 38

Content

Description

This slide, likely from an Artificial Intelligence Basics course focusing on image processing or signal processing, presents a solution strategy for processing signals or images, emphasizing the "smooth first" approach. The title "Solution: smooth first" clearly indicates the method. The slide displays four plots, each representing a different stage in the process. The top plot shows the original signal f . The second plot shows the kernel h , which is a Gaussian function (bell-shaped curve). The third plot shows the convolution of f and h , denoted as $f * h$. This convolution operation is a smoothing operation, effectively reducing high-frequency noise in the signal. The bottom plot shows the derivative of the smoothed signal, $\frac{d}{dx}(f * h)$. The value of $\sigma = 50$ is shown, indicating the standard deviation of the Gaussian kernel used for smoothing. The horizontal axis in all plots represents time or spatial position, and the vertical axis represents the signal intensity. The plots are labeled with their respective stages (Signal, Kernel, Convolution, Differentiation). The source is credited to S. Seitz, suggesting a specific author or researcher associated with the material. The context suggests that the slide is part of a discussion on image noise, its effects on image processing tasks, and techniques for noise reduction, particularly edge detection, using a Gaussian smoothing filter before differentiation.

Figures



Slide 39

Content

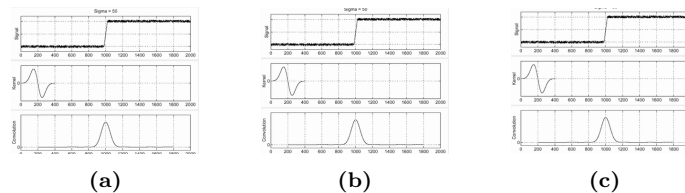
Associative property of convolution

Differentiation is a convolution Convolution is associative This saves us one operation: $\frac{d}{dx}(f * h) = f * \frac{d}{dx}h$

Description

This slide, likely from an Artificial Intelligence Basics course focusing on image processing or signal processing, introduces the associative property of convolution in the context of differentiation. The title "Associative property of convolution" clearly states the topic. The text explains that differentiation is a convolution operation, and convolution is associative. This means that the derivative of a convolution of two functions (f and h) is equivalent to convolving the first function with the derivative of the second function. The equation $\frac{d}{dx}(f * h) = f * \frac{d}{dx}h$ formally expresses this associative property. Below the equation, there are three plots. The top plot shows a signal $f(x)$. The middle plot shows a kernel $h(x)$, which is likely a Gaussian function used for smoothing. The bottom plot shows the convolution of $f(x)$ and $\frac{d}{dx}h(x)$, which is the derivative of the kernel convolved with the signal. The plots are labeled with "Signal", "Kernel", and "Convolution". The text "Sigma = 50" indicates the standard deviation of the Gaussian kernel used for smoothing. The horizontal axis represents time or spatial position, and the vertical axis represents the signal intensity. The source is credited to S. Seitz, indicating a specific author or researcher associated with the material. The context suggests that the slide is part of a discussion on image noise, its effects on image processing tasks, and techniques for noise reduction, particularly edge detection, using the associative property of convolution to efficiently compute the derivative of a smoothed signal.

Figures



Slide 40

Content

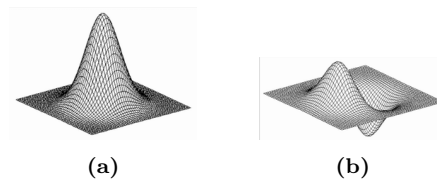
2D edge detection filters

- Gaussian $h_{\sigma}(u, v) = \frac{1}{2\pi\sigma^2} e^{-\frac{u^2+v^2}{2\sigma^2}}$
- derivative of Gaussian (x) $\frac{\partial}{\partial x} h_{\sigma}(u, v)$

Description

This slide from an Artificial Intelligence Basics course, likely focusing on image processing, presents two 2D filters crucial for edge detection. The title "2D edge detection filters" clearly indicates the topic. The slide displays a Gaussian function and its derivative with respect to x, both represented as 2D surfaces. The Gaussian function, $h_{\sigma}(u, v)$, is defined mathematically as $h_{\sigma}(u, v) = \frac{1}{2\pi\sigma^2} e^{-\frac{u^2+v^2}{2\sigma^2}}$. This is a 2D Gaussian kernel, commonly used for smoothing images before edge detection. The second filter is the derivative of the Gaussian with respect to x, denoted as $\frac{\partial}{\partial x} h_{\sigma}(u, v)$. This derivative highlights edges in the image. The parameter σ controls the width of the Gaussian kernel, influencing the degree of smoothing and the sensitivity to edges. The context suggests that the slide is part of a discussion on image processing techniques, specifically edge detection using Gaussian filters, and the mathematical formulation of these filters.

Figures



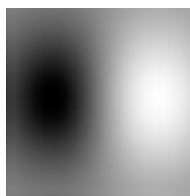
Slide 41

Content

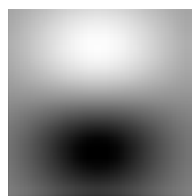
Description

This slide, likely from an Artificial Intelligence Basics course focusing on image processing, displays the concept of a derivative of a Gaussian filter. The title "Derivative of Gaussian filter" clearly indicates the topic. Two 3D surface plots are shown, representing the Gaussian function in 2D space. These plots visually depict the Gaussian filter's shape, which is a bell curve. The plots are likely showing the Gaussian function and its derivative in the x-direction. The x-axis and y-axis are labeled, and the z-axis represents the intensity or amplitude of the function. The color gradient on the surface plots indicates the varying values of the function. Below the 3D plots, two grayscale images are shown. These images represent the application of the Gaussian filter and its derivative to a sample image. The left image likely shows the result of applying the Gaussian filter, which smooths the image. The right image likely shows the result of applying the derivative of the Gaussian filter, which highlights edges and changes in intensity in the image. The context suggests that the slide is part of a discussion on image processing techniques, specifically edge detection using Gaussian filters, and the visual representation of the Gaussian filter and its derivative.

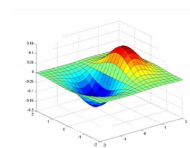
Figures



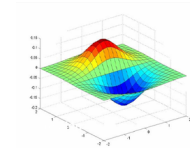
(a)



(b)



(c)



(d)

Slide 42

Content

FIGURE 5.3: The scale (i.e., σ) of the Gaussian used in a derivative of Gaussian filter has significant effects on the results. The three images show estimates of the derivative in the x direction of an image of the head of a zebra obtained using a derivative of Gaussian filter with σ of one pixel, three pixels, and seven pixels (left to right). Note how images at a finer scale show some hair, the animal's whiskers disappear at a medium scale, and the fine stripes at the top of the muzzle disappear at the coarser scale.

Description

This slide, likely from an Artificial Intelligence Basics course on image processing, demonstrates the impact of the Gaussian filter's scale parameter (σ) on edge detection results. It shows three images of a zebra's head, progressively processed with a derivative of Gaussian filter using different kernel sizes (one pixel, three pixels, and seven pixels). The images are presented from left to right, corresponding to increasing values of σ . The description highlights how the finer details (hair) are visible at the smaller σ (one pixel), while the whiskers (medium-scale details) disappear at a medium σ (three pixels), and the fine stripes at the top of the muzzle vanish at the larger σ (seven pixels). This illustrates how the scale of the Gaussian filter directly affects the level of smoothing and the detection of edges and features at different scales in an image. The context suggests a discussion on image filtering, edge detection, and the importance of choosing an appropriate filter scale for specific image analysis tasks.

Figures

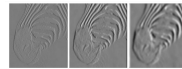


FIGURE 5.3 The scale (i.e., σ) of the Gaussian used in a derivative of Gaussian filter has significant effects on the results. The three images show estimates of the derivative in the x direction of an image of the head of a zebra obtained using a derivative of Gaussian filter with σ of one pixel, three pixels, and seven pixels (left to right). Note how images at a finer scale show some hair, the animal's whiskers disappear at a medium scale, and the fine stripes at the top of the muzzle disappear at the coarser scale.

(a)

Slide 43

Content

Two Dimensional Gaussian

- Anisotropic: $G_{\sigma_x, \sigma_y}(x, y) = \frac{1}{2\pi\sigma_x\sigma_y} \cdot e^{-\frac{1}{2}\left(\frac{x^2}{\sigma_x^2} + \frac{y^2}{\sigma_y^2}\right)}$
- Isotropic: $G_{\sigma}(x, y) = \frac{1}{2\pi\sigma^2} \cdot e^{-\frac{1}{2\sigma^2}(x^2+y^2)}$

Description

This slide from an Artificial Intelligence Basics course, likely focusing on image processing, defines two types of two-dimensional Gaussian functions. The title "Two Dimensional Gaussian" clearly indicates the topic. The slide presents the mathematical formulas for both an anisotropic and an isotropic Gaussian function.

The anisotropic Gaussian, $G_{\sigma_x, \sigma_y}(x, y)$, is defined as $G_{\sigma_x, \sigma_y}(x, y) = \frac{1}{2\pi\sigma_x\sigma_y} \cdot e^{-\frac{1}{2}\left(\frac{x^2}{\sigma_x^2} + \frac{y^2}{\sigma_y^2}\right)}$. This function has different standard deviations (σ_x and σ_y) along the x and y axes, resulting in an elliptical shape. The isotropic Gaussian, $G_{\sigma}(x, y)$, is defined as $G_{\sigma}(x, y) = \frac{1}{2\pi\sigma^2} \cdot e^{-\frac{1}{2\sigma^2}(x^2+y^2)}$. This function has the same standard deviation (σ) along both axes, resulting in a circular shape. The parameter σ (or σ_x and σ_y) controls the width of the Gaussian function, influencing the degree of smoothing. The context suggests that the slide is part of a discussion on image processing techniques, specifically the use of Gaussian functions for smoothing and filtering images.

Figures